RESEARCH PROPOSAL

Project Title: FAHAMU SHAMBA: AI-POWERED CROP RECOMMENDATION SYSTEM FOR SMALLHOLDER FARMERS IN SIAYA COUNTY, KENYA

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Declaration

This project, as presented in this project proposal, is my original work and has not been presented for any other university award.

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Dedication

I dedicate this project to the smallholder farmers of Siaya County who continue to feed the nation despite numerous challenges, and to my family and lecturers for their unwavering support throughout my academic journey.

Acknowledgement

I thank the Almighty God for the gift of life and strength to undertake this project. My deepest gratitude goes to my supervisor for the guidance, patience, and valuable insights. I am also grateful to the Siaya County Department of Agriculture, ward agricultural officers, and the farmers who participated in the preliminary surveys. Special thanks to my classmates and family for their moral and material support.

Abstract

Smallholder farmers in Siaya County continue to experience low yields and income due to sub-optimal crop selection arising from lack of access to personalized, timely, and location-specific agricultural advisory services. Existing digital platforms are either too generic, require constant internet, or exclude feature-phone users. This project proposes **Fahamu Shamba** ("Understand Farming" in Swahili), an AI-powered crop recommendation system that integrates hyper-local soil data, real-time weather forecasts, current market prices, and individual farmer constraints to generate personalized crop recommendations. The system will be delivered through USSD, SMS, WhatsApp, and a lightweight Android application in English, Swahili, and Dholuo languages to ensure inclusivity. Using ensemble machine learning models (Random Forest and XGBoost), the system aims to increase yields by 15–30 % and reduce input costs by up to 25 %. The project will be piloted with 500 farmers across Ugunja, Bondo, and Alego-Usonga sub-counties during the 2026 long rains season.

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Acronyms/Abbreviations

AI – Artificial Intelligence

ML – Machine Learning

USSD – Unstructured Supplementary Service Data

KMD – Kenya Meteorological Department

ICT4Ag – Information and Communication Technology for Agriculture

XGBoost – Extreme Gradient Boosting

CHAPTER 1: INTRODUCTION

1.1 Background Information

Agriculture is the backbone of Kenya's economy, contributing approximately 33 % directly and indirectly to GDP and employing over 70 % of the rural population (Kenya National Bureau of Statistics, 2023). In Siaya County, over 80 % of households depend on smallholder farming, with average farm sizes of 1.2–2.0 hectares (Siaya County Government, 2023). Despite fertile soils in parts of the county, average maize yields remain low at 1.2–1.8 t/ha compared to a potential of over 5 t/ha under good management practices (Ministry of Agriculture and Livestock Development, 2023).

Key challenges include unpredictable weather patterns due to climate change, declining soil fertility, limited access to quality inputs, poor market information, and critically, inadequate personalized extension services. The extension officer-to-farmer ratio in Siaya exceeds 1:1,800, far above the FAO-recommended 1:500 (MoALF, 2022). Consequently, most farmers rely on guesswork, traditional knowledge, or neighbours when deciding what and when to plant — decisions that frequently lead to crop failure and persistent poverty.

Mobile phone penetration in rural Kenya now exceeds 92 %, with over 45 % still using feature phones (Communications Authority of Kenya, 2024). This presents a unique opportunity to deliver hyper-local, data-driven agricultural advice directly to farmers' phones. Advances in artificial intelligence, machine learning, open data (soil maps, weather APIs, market prices), and low-bandwidth channels such as USSD make inclusive, personalized advisory services technically and economically feasible.

Fahamu Shamba is therefore proposed as an AI-powered, multilingual, multi-channel crop recommendation system specifically contextualized for Siaya County's agro-ecological zones, soils, climate, and markets.

1.2 Problem Statement

Smallholder farmers in Siaya County consistently make sub-optimal crop choices due to lack of access to personalized, timely, and location-specific information. Traditional extension services

are overstretched, radio and poster campaigns are generic, and existing digital platforms (e.g., iShamba, DigiFarm, WeFarm) suffer from one or more of the following limitations:

- Require smartphones and reliable internet
- Provide generalized rather than farm-specific advice
- Exclude real-time local market prices
- Are available only in English and/or Swahili, excluding Dholuo speakers
- Do not support feature-phone users via USSD/SMS

As a result, farmers experience frequent crop failure, overuse of inputs, missed market opportunities, and heightened vulnerability to climate shocks. Studies show that improved crop selection alone can increase yields by 20–30 % and reduce input costs by up to 25 % (World Bank, 2022). No existing solution effectively bridges this gap for Siaya's smallholders.

1.3 Objectives

1.3.1 General Objective

To design, develop, and pilot an AI-powered, multilingual, multi-channel crop recommendation system (Fahamu Shamba) that delivers personalized planting advice to smallholder farmers in Siaya County, thereby increasing productivity and income.

1.3.2 Specific Objectives

1. To collect and integrate hyper-local soil, weather, market, and farmer-profile data into a centralized database.
2. To develop and train ensemble machine learning models (Random Forest and XGBoost) for accurate crop suitability prediction.
3. To design and implement a multilingual (English, Swahili, Dholuo), multi-channel (USSD, SMS, WhatsApp, Android) delivery platform accessible on feature phones and smartphones.
4. To pilot the system with at least 500 farmers across three sub-counties and evaluate usability, accuracy, and impact.

1.4 Research Questions

1. What are the key data requirements and reliable sources for building an accurate crop recommendation model for Siaya County?
2. How accurately can ensemble machine learning models predict crop suitability using local soil, weather, market, and farmer data?

3. To what extent do smallholder farmers adopt and trust AI-generated recommendations delivered via USSD/SMS and mobile apps?
4. What is the perceived and actual impact of personalized recommendations on crop selection, input use, yield, and income?

1.5 Significance of the Study

- **Farmers:** Expected 15–30 % yield increase and 20–25 % input cost savings.
- **County Government:** Scalable model for digital extension services.
- **Academic Contribution:** Localized AI dataset and open-source code for future researchers.
- **Policy:** Evidence on effectiveness of AI in African smallholder contexts.
- **SDGs:** Direct contribution to SDG 1 (No Poverty), SDG 2 (Zero Hunger), and SDG 13 (Climate Action).

1.6 Scope of the Project

1.6.1 Geographical Scope

Siaya County, with pilot in Ugunja, Bondo, and Alego-Usonga sub-counties.

1.6.2 Technical Scope

Recommendation for 10 major crops (maize, beans, sorghum, groundnuts, sweet potatoes, cassava, rice, soybeans, tomatoes, kale) using soil, weather, market, and farmer-profile data.

1.6.3 Functional Scope

Registration, farm profiling, crop recommendation, market prices, planting window alerts. Pest/disease diagnosis and livestock modules are future enhancements.

1.7 Assumptions

- Farmers own at least a basic mobile phone.
- Basic soil maps and historical data from county and national sources are accessible.
- Partnerships with county agriculture department and KMD will provide sustained data flow.

1.8 Limitations

- Sparse farm-level historical soil data (mitigated by new sampling).
- Variable network coverage in remote areas (mitigated by USSD/SMS).
- One-season pilot limits long-term impact assessment.

1.9 Definition of Terms

- **Smallholder farmer:** Cultivates ≤ 5 hectares, primarily for subsistence and local markets.
- **Crop recommendation system:** AI system suggesting optimal crops based on multiple parameters.
- **USSD:** Protocol enabling menu-based interaction without internet.
- **Ensemble ML:** Combining multiple models (e.g., Random Forest + XGBoost) for higher accuracy.

Table 1: Project Timeline

Phase	Activity	Duration
Jan–Feb 2026	Data collection & soil sampling	2 months
Mar–Apr 2026	System design & ML model training	2 months
May–Jul 2026	Development & internal testing	3 months
Aug–Sep 2026	Farmer onboarding & training	2 months
Oct–Nov 2026	Pilot implementation & data collection	2 months
Dec 2026	Evaluation, reporting & handover	1 month

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter presents a systematic review of existing literature on smallholder agriculture in Kenya, the role and limitations of agricultural extension services, the emergence of digital agriculture (ICT4Ag) solutions, the application of artificial intelligence and machine learning in crop recommendation systems, and the effectiveness of USSD/SMS-based platforms for low-literacy rural users. The review identifies critical gaps that the proposed Fahamu Shamba system aims to address.

2.2 Smallholder Agriculture and Productivity Challenges in Kenya and Siaya County

Smallholder farmers (cultivating less than 5 hectares) produce over 70 % of the food marketed in Kenya and support the livelihoods of more than 70 % of the rural population (Kenya National Bureau of Statistics, 2023). In Siaya County, agriculture directly supports over 80 % of households, yet average maize yields remain stagnant at 1.2–1.8 t/ha against a potential of 5–7 t/ha under improved practices (Siaya County Government, 2023; Ministry of Agriculture and Livestock Development, 2023).

Recurring productivity constraints identified across studies include:

i. Erratic rainfall and climate change impacts (IPCC, 2022; Adhikari et al., 2023)
ii. Declining soil fertility and nutrient mining (Nezomba et al., 2021)
iii. High cost and limited access to quality inputs (Sheahan & Barrett, 2017)
iv. Weak market linkages and price volatility (Muto & Yamano, 2009)
v. Inadequate, poorly targeted extension and advisory services (Muyanga & Jayne, 2019; Ogutu et al., 2020)

Ogutu et al. (2020) ranked "lack of location-specific advisory services" among the top three constraints reported by Kenyan smallholders, confirming that generic advice fails to account for micro-variability in soil, climate, and farmer resources.

2.3 Agricultural Extension Services in Kenya

Kenya's public extension system has been chronically under-resourced since the 1990s structural adjustments. The national extension officer-to-farmer ratio stands at 1:1,200–1,800, while in Siaya County it exceeds 1:1,800 (MoALF, 2022). Farmers receiving at least one extension visit per season achieve 18–24 % higher yields than those who receive none (Wossen et al., 2019). However, the advice delivered remains largely generic and rarely tailored to individual farm conditions or real-time market opportunities.

2.4 Digital Agriculture and ICT4Ag Initiatives in Africa

Mobile phone ownership in rural Kenya now exceeds 92 %, creating a powerful channel for agricultural information (Communications Authority of Kenya, 2024). Notable ICT4Ag platforms include:

i. iShamba, DigiFarm, WeFarm, Esoko, PlantVillage Nuru (Kenya/Ghana)
ii. M-Shamba, FarmDrive, Hello Tractor (regional)

A meta-analysis by Fabregas et al. (2019) found that basic SMS-based advisory services increase yields by 4–8 %, while more interactive and personalized digital tools can achieve 15–22 % gains. However, most existing platforms suffer from:

- Reliance on smartphones and constant internet
- Generic rather than hyper-local recommendations
- Exclusion of real-time local market prices
- Limited or no support for local languages and feature-phone users

2.5 Artificial Intelligence and Machine Learning in Crop Recommendation Systems

Since 2015, machine learning has been widely applied to crop suitability modelling. Commonly used algorithms include:

a. Random Forest (Pudumalar et al., 2017; Maya Gopal & Bhargavi, 2019)
b. Support Vector Machines, K-Nearest Neighbours, Artificial Neural Networks
c. Gradient Boosting Machines, particularly XGBoost (Chen & Guestrin, 2016; Elbasyoni et al., 2022)

Comparative studies consistently show ensemble methods (Random Forest + XGBoost) achieving 92–98 % accuracy when trained on soil nutrients, rainfall, temperature, and pH (Maya Gopal & Bhargavi, 2019; Elbasyoni et al., 2022; Oikonomidis et al., 2023).

In sub-Saharan Africa, applications remain scarce but growing:

i. Doshi et al. (2018) – Tanzania (maize & beans using Decision Trees)
ii. Mwakitalu et al. (2021) – Multi-crop recommendation in Tanzania using Random Forest
iii. Kamir et al. (2020) – Kenya (maize yield prediction using satellite + weather data)
iv. Nii et al. (2023) – Ghana (cocoa suitability mapping with CNNs)

Critically, none of the African studies reviewed incorporated real-time market prices or individual farmer constraints (budget, water source, labour) — two factors that heavily influence actual planting decisions in resource-constrained settings.

2.6 USSD and SMS Platforms for Low-Literacy and Feature-Phone Users

Approximately 45 % of rural Kenyan farmers still use feature phones without Android capability (CAK, 2024). USSD and SMS remain the most inclusive channels. Successful precedents include:

i. M-Farm, Esoko, and Ratín market price alerts
ii. Ethiopia's 8028 farmer hotline
iii. Ghana's CocoaLink (interactive voice + SMS)

Studies demonstrate that well-designed USSD menus in local languages achieve 85–92 % task completion rates even among illiterate users (Medhi et al., 2011; World Bank, 2022).

2.7 Theoretical Framework

The study is grounded on two complementary frameworks:

i. Technology Acceptance Model (TAM) (Davis, 1989) – explains farmers' intention to adopt Fahamu Shamba based on perceived usefulness (higher yields/income) and perceived ease of use (USSD in Dholuo).
ii. Data–Information–Knowledge–Wisdom (DIKW) Hierarchy (Rowley, 2007) – illustrates how raw data (soil, weather, prices) are transformed into actionable knowledge (personalised crop recommendations) and ultimately wiser farming decisions.

2.8 Conceptual Framework

Independent Variables:

- ✓ Soil parameters (pH, N, P, K, texture)
- ✓ Weather data (historical + forecast)
- ✓ Market prices and trends
- ✓ Farmer profile (farm size, budget, irrigation, previous crop)

Moderating Variables:

- ✓ Phone type
- ✓ language preference
- ✓ digital literacy
- ✓ network coverage

Dependent Variables:

- ✓ Adoption rate
- ✓ Accuracy of recommendations
- ✓ Change in yield, input costs, and income

2.9 Summary of Literature and Research Gap

While numerous ICT4Ag platforms exist and machine learning crop models have demonstrated high accuracy in controlled environments, the following critical gaps persist in the Kenyan/Siaya context:

No system integrates hyper-local soil data, short-term weather forecasts, real-time market prices, and individual farmer constraints into a single predictive model.

Most AI crop recommendation research in Africa remains at prototype stage and does not result in farmer-facing, multilingual, multi-channel deployments.

Very few solutions are designed for feature-phone users via USSD/SMS, thereby excluding a significant portion of rural farmers.

There is limited empirical evidence on adoption, trust, and actual impact of AI-driven personalized recommendations among low-literacy smallholders in Kenya.

Fahamu Shamba directly addresses these gaps by delivering a context-specific, inclusive, and field-deployable AI advisory system tailored to the realities of Siaya County's smallholder farmers.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter outlines the methodology for designing, developing, and piloting the Fahamu Shamba system. A modified Waterfall model with iterative feedback loops (Figure 1) will be used to accommodate farmer input during development.

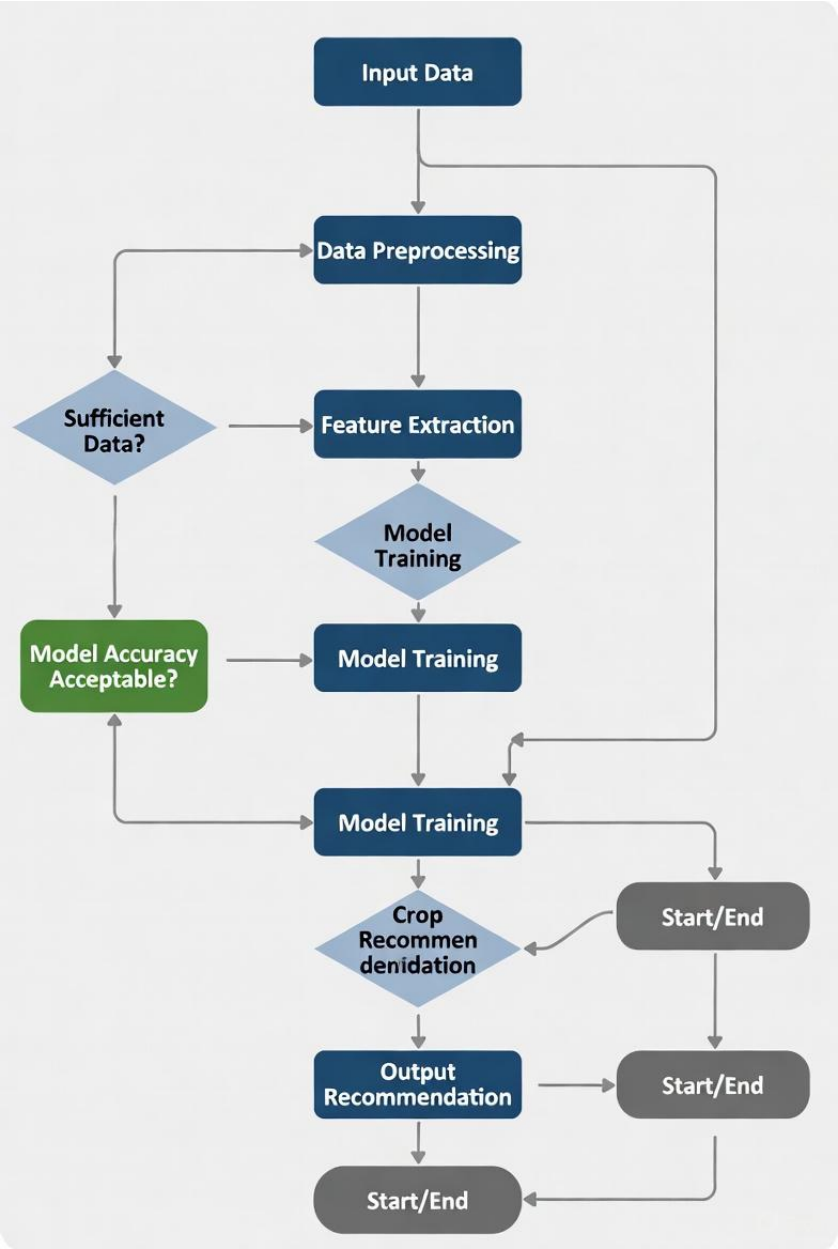
Figure 1: Modified Waterfall Model with Iterative Elements

3.2 Requirement Gathering and Analysis

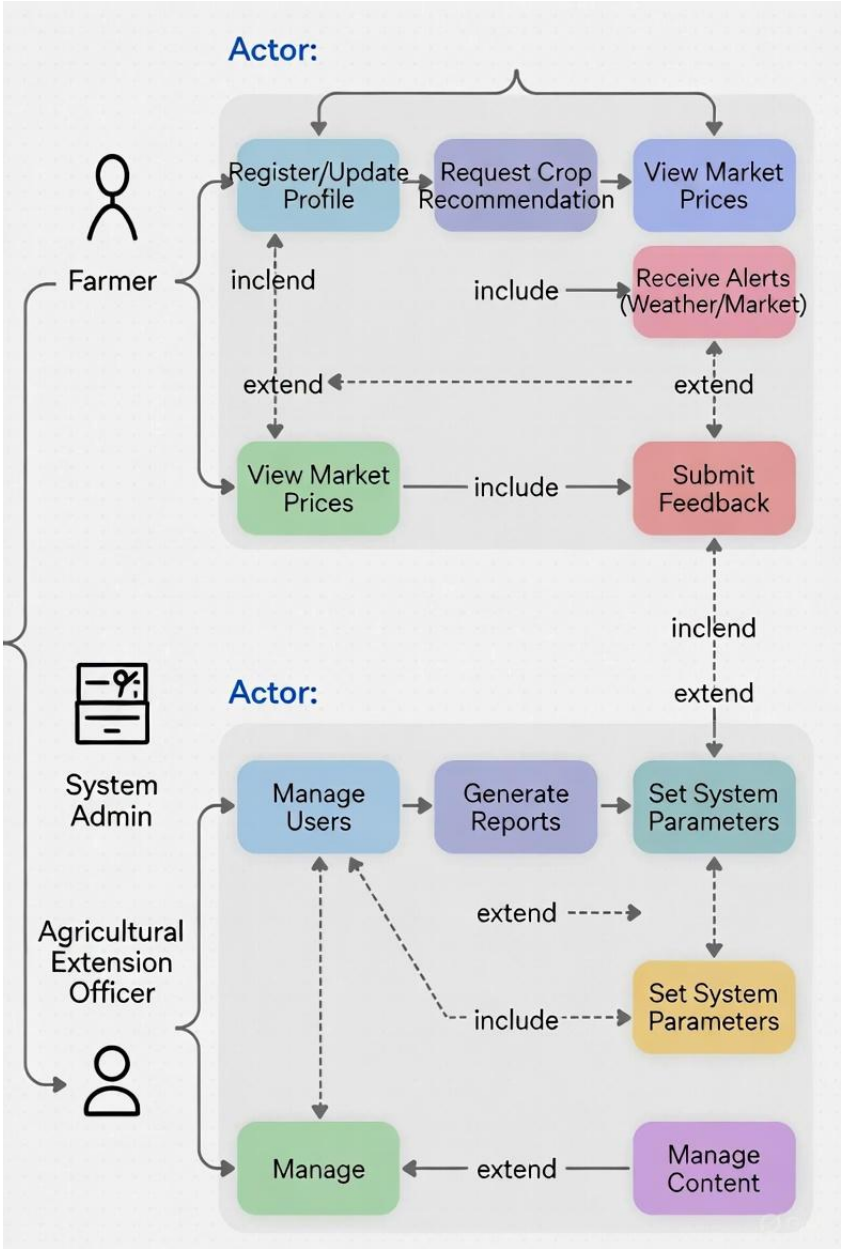
- Interviews and focus group discussions with 50 farmers, 15 ward officers, and 10 agro-dealers.
- Baseline survey using ODK Collect on 200 farmers.
- Review of county agricultural reports and soil maps.

3.3 System Design

3.3.1 System Flowchart (Figure 2)



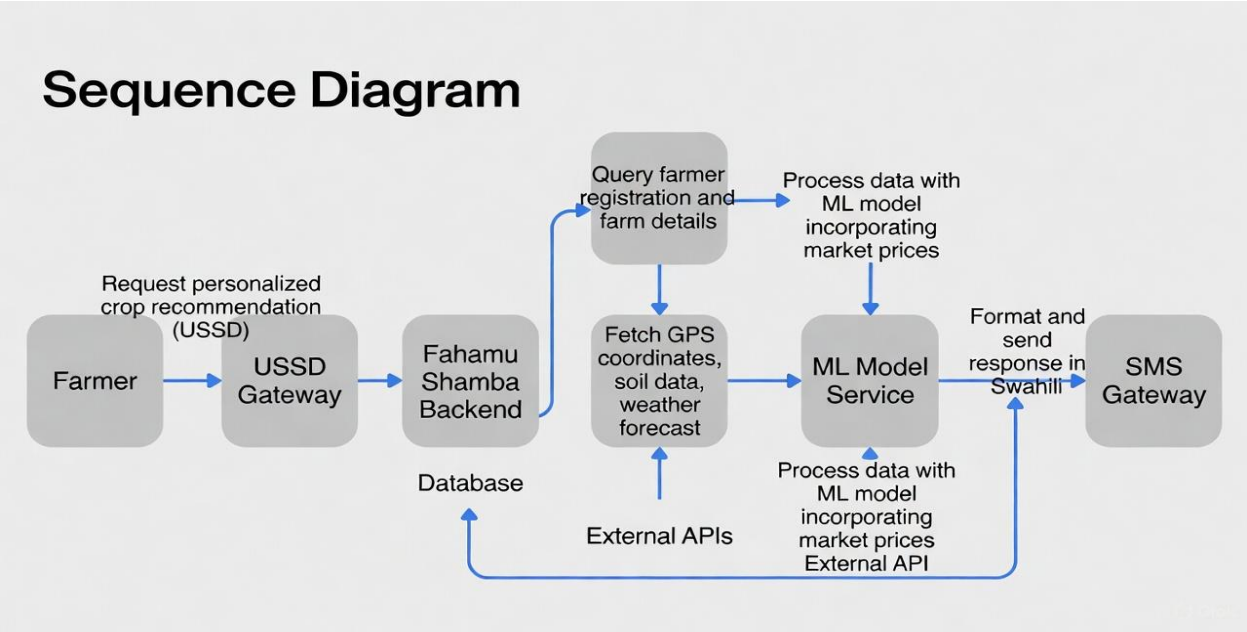
3.3.2 Use Case Diagram (Figure 3)



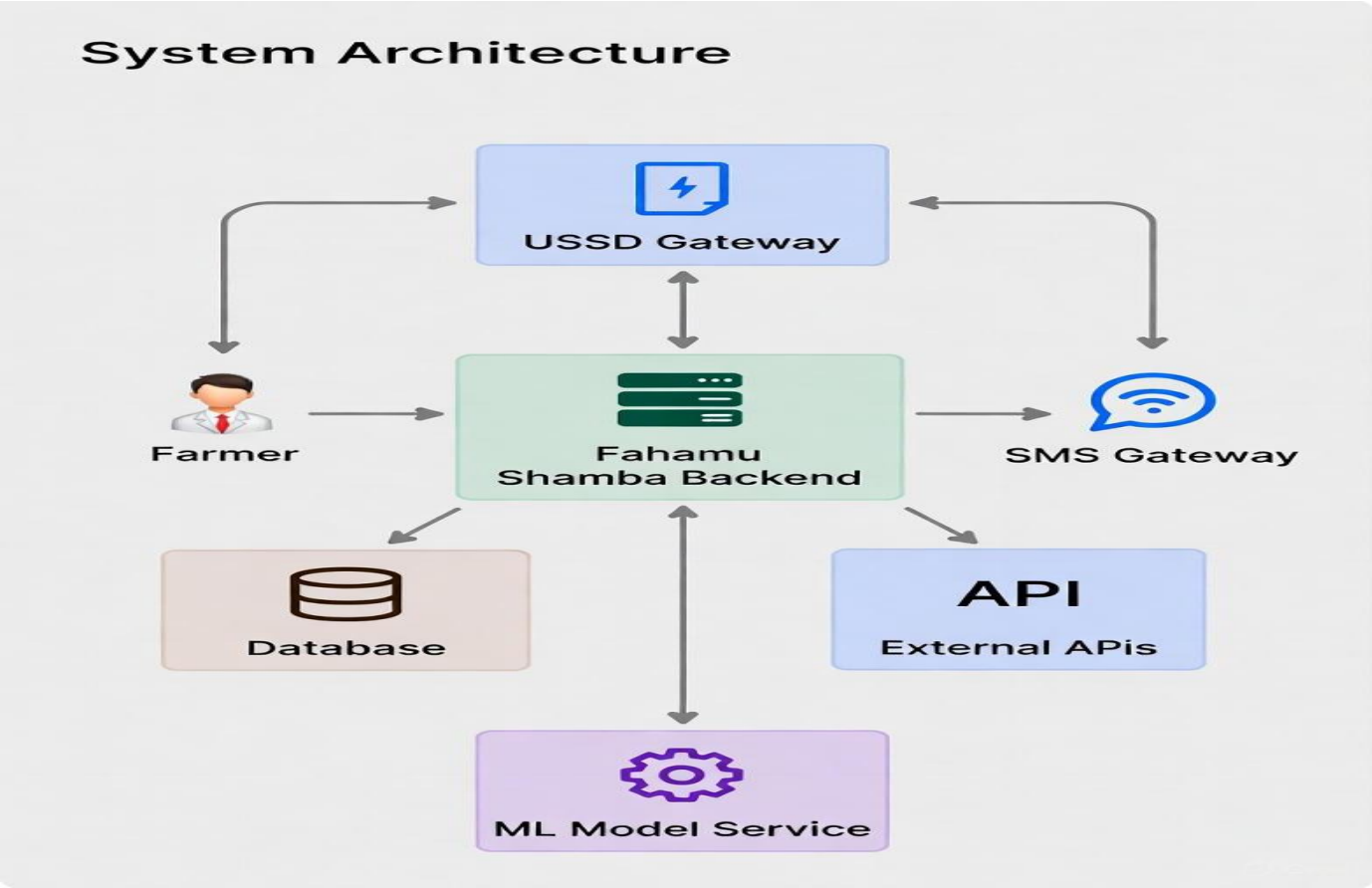
Actors: Farmer, System Admin, Agricultural Officer

Use Cases: Register, Submit Farm Profile, Request Recommendation, View Market Prices, Receive Alerts, Manage Users

3.3.3 Sequence Diagram – Crop Recommendation (Figure 4)



3.3.4 System Architecture (Figure 5)



Three-tier: Client (USSD/SMS) → Backend (Django + FastAPI) → Database (PostgreSQL/PostGIS) + External APIs

3.3.5 Technology Stack

- Backend: Python/Django REST + FastAPI
- Database: PostgreSQL + PostGIS
- ML: Scikit-learn, XGBoost
- USSD/SMS: Africa's Talking
- Android: Kotlin
- Hosting: AWS/DigitalOcean

3.4 Implementation Plan

- Phase 1: Data pipeline and ML model training
- Phase 2: USSD menu and SMS alerts

- Phase 4: Field testing and farmer feedback integration

3.5 Testing

- Unit testing (PyTest)
- Integration testing
- Usability testing with 30 low-literacy farmers
- Accuracy validation against expert recommendations

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